

15.20 The 250-mm-radius circular plate shown is initially at rest and has an angular acceleration defined by the relation $\alpha = \alpha_0 \cos(\pi t/T)$. Knowing that $T = 1.5$ s and $\alpha_0 = 10$ rad/s², determine the magnitude of the total acceleration of point B when (a) $t = 0$, (b) $t = 0.5$ s, (c) $t = 0.75$ s.

Grade: 4/4

Excellent!

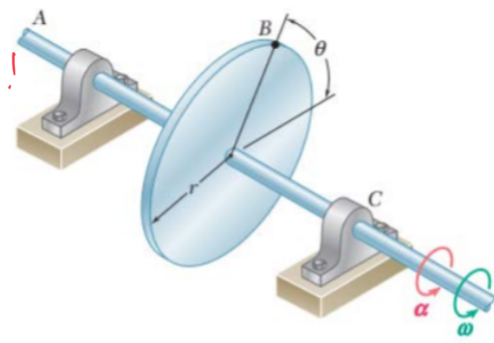


Fig. P15.18, P15.19, and P15.20

$$\alpha = \alpha_0 \cos\left(\frac{\pi t}{T}\right)$$

$$\Rightarrow \omega = \int_0^t \alpha_0 \cos\left(\frac{\pi t}{T}\right) dt$$

$$= \frac{T}{\pi} \alpha_0 \sin\left(\frac{\pi t}{T}\right)$$

a) When $t = 0$,

$$a_t = r\alpha$$

$$= (250 \times 10^{-3})(10)$$

$$= 2.5 \text{ m/s}^2$$

$$a_n = r\omega^2$$

$$= (250 \times 10^{-3})(0)$$

$$= 0 \text{ m/s}^2$$

$$a_B = \sqrt{a_t^2 + a_n^2} = 2.5 \text{ m/s}^2$$

b) When $t = 0.5$,

$$a_t = (250 \times 10^{-3}) \left(10 \cos \frac{0.5\pi}{1.5} \right)$$

$$= 1.25 \text{ m/s}^2$$

$$a_n = r\omega^2$$

$$= (250 \times 10^{-3}) \left[\frac{1.5}{\pi} \times 10 \sin\left(\frac{0.5\pi}{1.5}\right) \right]^2$$

$$= 5.7 \times \sin^2\left(\frac{\pi}{3}\right)$$

$$= 4.275 \text{ m/s}^2$$

$$a_B = \sqrt{a_t^2 + a_n^2} = \sqrt{1.25^2 + 4.275^2} = 4.45 \text{ m/s}^2$$

c) When $t = 0.75$,

$$a_t = (250 \times 10^{-3}) \left(10 \cos \frac{0.75\pi}{1.5} \right)$$

$$= 0 \text{ m/s}^2$$

$$a_n = (250 \times 10^{-3}) \left[\frac{1.5}{\pi} \times 10 \sin\left(\frac{0.75\pi}{1.5}\right) \right]^2$$

$$= 5.7 \times \sin^2\left(\frac{\pi}{2}\right)$$

$$= 5.7 \text{ m/s}^2$$

$$a_B = \sqrt{0^2 + 5.7^2} = 5.7 \text{ m/s}^2$$

15.28 Cylinder A is moving downward with a velocity of 9 ft/s when the brake is suddenly applied to the drum. Knowing that the cylinder moves 18 ft downward before coming to rest and assuming uniformly accelerated motion, determine (a) the angular acceleration of the drum, (b) the time required for the cylinder to come to rest.

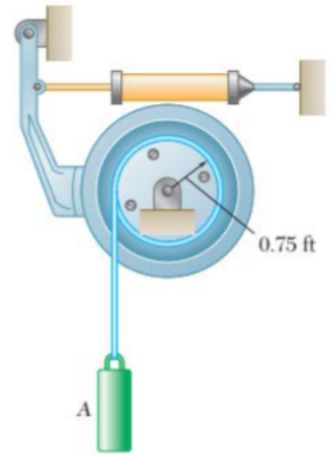


Fig. P15.28 and P15.29

$$V_0 = 9 \text{ ft/s} \quad r = 0.75 \text{ ft}$$

considering cylinder A:

$$v^2 = v_0^2 + 2as$$

$$0 = 9^2 + 2a(18)$$

$$a = -2.25 \text{ ft/s}^2, \uparrow$$

considering drum:

$$V_0 = r\omega$$

$$9 = 0.75\omega$$

$$\omega = 12 \text{ rad/s}$$

a)

$$a_t = r\alpha$$

$$-2.25 = 0.75\alpha$$

$$\alpha = -3 \text{ rad/s}^2, \text{ clockwise}$$

b) when $t = t_1$ $\omega = 0$

$$\omega = \omega_0 + \alpha t$$

$$0 = 12 + (-3)t_1$$

$$t_1 = 4 \text{ s}$$

15.40 Small wheels have been attached to the ends of rod AB and roll freely along the surfaces shown. Knowing that wheel A moves to the left with a constant velocity of 1.5 m/s, determine (a) the angular velocity of the rod, (b) the velocity of end B of the rod.

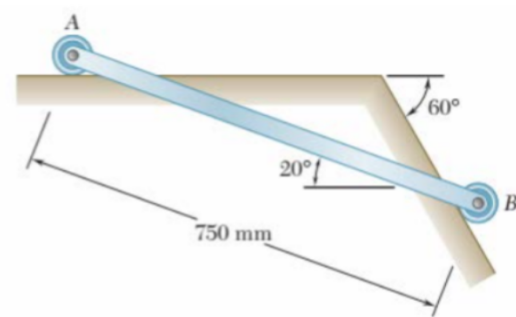
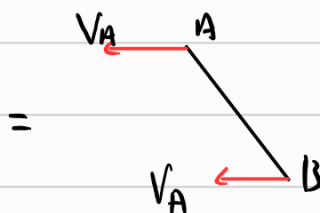
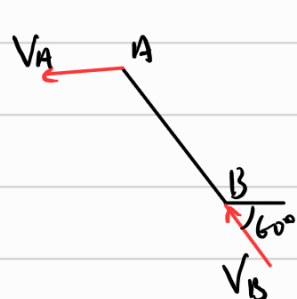
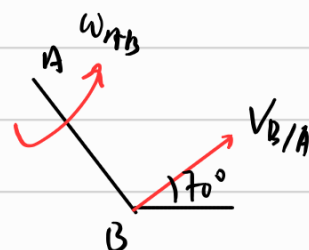


Fig. P15.40

$$V_A = 1.5 \text{ m/s}$$

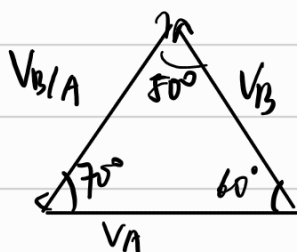


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Nice drawing!

$$V_B = V_A + V_{B/A}$$



By sine rule,

$$\frac{V_A}{\sin 50^\circ} = \frac{V_{B/A}}{\sin 60^\circ}$$

$$V_{B/A} = \frac{1.5}{\sin 50^\circ} \sin 60^\circ = 1.696 \text{ m/s}$$

a) $v = r\omega$

$$\omega_{AB} = \frac{v}{r} = \frac{1.696}{0.75} = 2.261 \text{ rad/s}$$

$$b) \frac{V_B}{\sin 70^\circ} = \frac{V_A}{\sin 50^\circ}$$

$$V_B = 1.5 \left(\frac{\sin 70^\circ}{\sin 50^\circ} \right) = 1.84 \text{ m/s} \quad \angle 60^\circ$$

15.50 Gear A rotates with an angular velocity of 120 rpm clockwise. Knowing that the angular velocity of arm AB is 90 rpm clockwise, determine the corresponding angular velocity of gear B.

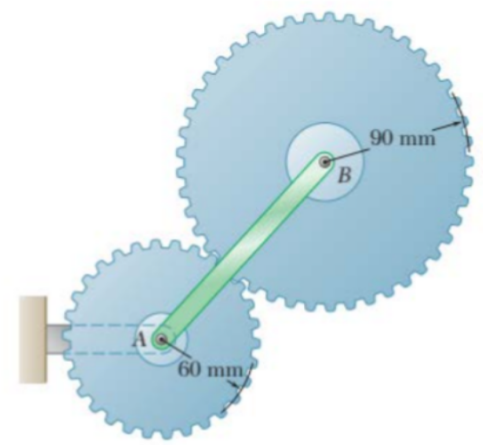
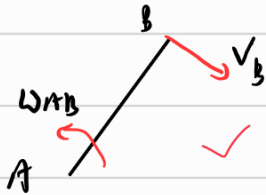
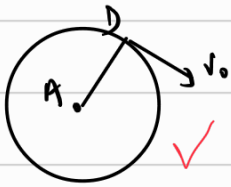


Fig. P15.50 and P15.51



$$\omega_{AB} = 90 \text{ rpm} \\ = \frac{398}{10} \times 2\pi = 3\pi \text{ rad/s} \downarrow$$



$$v_D = r_{AB} \omega_{AB} = 15 (3\pi) = 45\pi \text{ cm/s} \downarrow$$

$$\text{Gear A: } \omega_A = 120 \text{ rpm} = \frac{40}{60} \times 2\pi = 4\pi \text{ rad/s}$$

$$v_D = r_A \omega_A = 6 (4\pi) = 24\pi \text{ cm/s} \downarrow$$

$$\text{Gear B: } v_D = v_B + v_{D/B} \quad \checkmark$$

$$24\pi \downarrow = 45\pi \downarrow + v_{D/B}$$

$$v_{D/B} = 21\pi \text{ cm/s} \uparrow$$

$$\omega_B = \frac{v_{D/B}}{r_B} = \frac{21\pi}{9} \text{ rad/s} \downarrow$$